

Cascades, Multivariate Extremes, and Transformers

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Introduction and Motivation

Vision

- **Generative AI** has been entering a new era. Many opportunities beyond text, images, video, etc.
- I believe it is important for us to position ourselves at the frontier of this shift. This is the natural next step.

Structure of today's talk

- Cascading Extremes.
- Generative Transformer-Based Approaches for Cascading Extremes.

Vision

A cutting-edge risk engine for **generative cascades of extremes**.

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RESEARCH



A Kolmogorov–Arnold Neural Model for Cascading Extremes

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Part I

Neural Statistical Modeling of Cascading Extremes

Introduction and Motivation

Compound, Cascading, and Complex Extreme Events

- While it is widely recognized by practitioners that **extreme events** tend to occur in **complex sequential forms** (Cutter2018; Raymond et al.2020), statistical modelling of such context from an EVT viewpoint is still underdeveloped.
- Multivariate EVT, though a natural starting point, falls short by:
 - disregarding the **triggering role** of certain events;
 - overlooking the order and sequential nature of **extreme event cascades**;
 - lacking the ability to model **feedback loops** between events.

Introduction and Motivation

The POC Surface

- Inspired by the multivariate EVT framework, in this talk I introduce a novel concept, the **POC (Probability of Cascade) surface**, to assess the probability of domino effects between extreme events conditionally on a covariate or feature vector $\mathbf{x} \in \mathbb{R}^p$.
- The proposed POC-based approach is fully general in the sense that the focus can be placed beyond the case where follow-up event is binary.
- In particular, we extend the framework to a **multi-class setting**, allowing for different types of follow-up extreme events.
- The case where the **follow-up event is continuous** includes as a particular case the conditional coefficient of extremal dependence introduced by [Lee et al.2024](#)).

Background

- To learn about the POC surface from the data, we develop a neural model grounded on **Kolmogorov's superposition theorem**. pause
- **Superpositions** are **functions of functions**.

Example (Superposition of univariate and bivariate functions)

$$f(x_1, x_2, x_3) = g(a(\alpha(x_1), \beta(x_2, x_3)), b(x_1, x_2)).$$

Theorem (Kolmogorov's superposition theorem)

Let $f : [0, 1]^d \rightarrow \mathbb{R}$ be a continuous function.

$$f(\mathbf{x}) = \sum_{i=1}^{2d+1} \phi_i^{(2)} \left(\sum_{j=1}^d \phi_{ij}^{(1)}(x_j) \right), \quad \mathbf{x} = (x_1, \dots, x_d)^T,$$

for some continuous one-dimensional functions $\phi_{ij}^{(1)}$ and $\phi_j^{(2)}$.



A. Kolmogorov

Background

- From an AI perspective, the theorem reveals a [two-layer neural network architecture](#) recently popularized by [Liu et al.2024](#)) following their extension to deeper settings.
- While [multi-layer perceptrons](#) are inspired by the [universal approximation theorem](#) (e.g., [Berlyand and Jabin2023](#)), [Kolmogorov–Arnold Networks \(KAN\)](#) are a novel and fast-evolving addition to the AI toolbox, and are rooted on Kolmogorov’s superposition theorem.
- A particularly impressive aspect of KAN is that they are based on the principle any multivariate continuous function can be expressed exactly using only $2d + 1$ outer functions and d inner functions.
- In addition to the many developments following [Liu et al.](#), it should be noted that other neural approaches based on this theorem had already appeared in the literature ([Lin and Unbehauen1993](#); [Sprecher and Draghici2002](#); [Montanelli and Yang2020](#); [Fakhoury et al.2022](#)).

Cascading Extremes

Modeling Chained Extreme Events

- Let $I = \{I_u : u \in \mathbb{R}\}$ be a Bernoulli process and $Y \sim F_Y$ be a continuous rv.
- We start by introducing the following functional, referred to as **alpha**, which plays a central role in our developments:

$$\alpha \equiv \alpha_I = \lim_{u \rightarrow y^*} P(I_u = 1 \mid Y > u).$$

- Notation: $y^* = \sup\{y : F_Y(y) < 1\}$ is the right endpoint of F_Y .
- Loosely, α is the probability of a **follow-up event** (like a tsunami $I_u = 1$), given a **trigger event** (such as an earthquake exceeding magnitude u).
- The nature of the Bernoulli process I defining the follow-up event opens up a variety of modeling possibilities as illustrated below.

Examples of Alpha

Example (Tail dependence coefficient)

If $I_u = I(Z > u)$, where Y and Z have common distribution, then

$$\alpha^{\text{TDC}} \equiv \alpha_I = \lim_{u \rightarrow y^*} P(Z > u \mid Y > u).$$

Thus, α includes the well-known tail dependence coefficient as a special case when the follow-up event involves Z being extreme and [observable](#). ■

Example (Extremal probabilistic index)

If $I_u = I(Z > Y)$, then

$$\alpha^{\text{PI}} \equiv \alpha_I = \lim_{u \rightarrow y^*} P(Z > Y \mid Y > u),$$

which can be regarded as extremal version of the [probabilistic index](#) (Thas et al.2012). ■

POC Surface

Setup and Definition

- Our setup keeps in mind that for some applications the variable Z in the above examples might be latent, but it assumes that the Bernoulli process I is always observable.
- In practice it is desirable to assess how the α functional may be impacted by a covariate or feature.

Definition (POC Surface)

Let $\mathbf{x} = (x_1, \dots, x_d)^T \in \mathcal{X} \subseteq \mathbb{R}^d$. The probability of cascade surface is defined as

$$\text{POC} = \{(\mathbf{x}, \alpha_I(\mathbf{x})) : \mathbf{x} \in \mathcal{X}\}, \quad \alpha_I(\mathbf{x}) = \lim_{u \rightarrow y^*} P(I_{u,\mathbf{x}} = 1 \mid Y_{\mathbf{x}} > u),$$

where $I = \{I_{u,\mathbf{x}} : (u, \mathbf{x}) \in \mathbb{R} \times \mathcal{X}\}$ is a random field with Bernoulli marginal distributions and $\{Y_{\mathbf{x}} : \mathbf{x} \in \mathcal{X}\}$ is a random field.

POC Surface

A Kolmogorov–Arnold Approach for Learning from Data

Starting Point (KAN):

$$\alpha_I(\mathbf{x}) = \sum_{i=1}^{2d+1} \phi_i^{(2)} \left(\sum_{j=1}^d \phi_{ij}^{(1)}(x_j) \right).$$

Or in function matrix notation

$$\alpha_I(\mathbf{x}) = (\Phi^{(2)} \circ \Phi^{(1)}) \mathbf{x},$$

where

$$\Phi^{(1)} = \begin{pmatrix} \phi_{1,1}^{(1)} & \cdots & \phi_{1,d}^{(1)} \\ \vdots & \ddots & \vdots \\ \phi_{2d+1,1}^{(1)} & \cdots & \phi_{2d+1,d}^{(1)} \end{pmatrix}, \quad \Phi^{(2)} = \begin{pmatrix} \phi_1^{(2)} \\ \vdots \\ \phi_{2d+1}^{(2)} \end{pmatrix}^T.$$

Issue: $\alpha_I(\mathbf{x})$ may not be in $[0, 1]$!

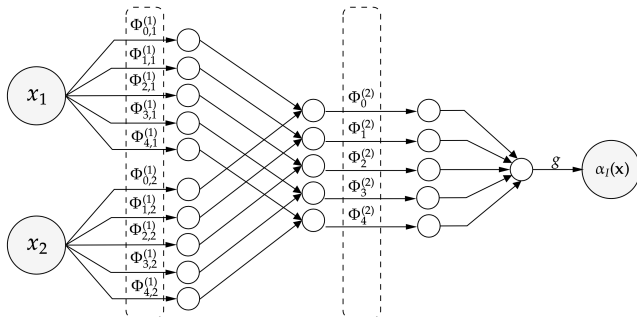
POC Surface

KANE: KAN with Natural Enforcement

Refined Version (KANE): Let $g : \mathbb{R} \rightarrow [0, 1]$, and set

$$\alpha_I(\mathbf{x}) = g \left(\sum_{i=1}^{2d+1} \phi_i^{(2)} \left(\sum_{j=1}^d \phi_{i,j}^{(1)}(x_j) \right) \right).$$

Or in function matrix notation $\alpha_I(\mathbf{x}) = g(\Phi^{(2)} \circ \Phi^{(1)}) \mathbf{x}$. **Now:** $\alpha_I(\mathbf{x}) \in [0, 1]$.



Deep POC Surface

A Deep Version of the Model *a la* Liu et al.

Deep Version (L Layer Model): Let $g : \mathbb{R} \rightarrow [0, 1]$, and set

$$\alpha_l(\mathbf{x}) = g(\Phi^{(L-1)} \circ \dots \circ \Phi^{(1)}),$$

where

$$\Phi^{(l)} = \begin{pmatrix} \Phi_{1,1}^{(l)} & \dots & \Phi_{1,n_l}^{(l)} \\ \vdots & \ddots & \vdots \\ \Phi_{n_l+1,1}^{(l)} & \dots & \Phi_{n_l+1,n_l}^{(l)} \end{pmatrix},$$

where n_l is the number of nodes in the l th layer.

Deep POC Surface

A Kolmogorov–Arnold Approach for Learning from Data

- Consider $m + 1$ equally-spaced knots, $t_0 < \dots < t_m$. We model the inner and outer functions as a linear combination of B-spline basis functions, that is,

$$\Phi_{i,j}^{(l)}(x) = \sum_{k=1}^K \beta_{i,j,k}^{(l)} B_k^p(x),$$

for $i = 1, \dots, d$, where $B_k^p(x)$ is a B-spline basis function of degree p evaluated at x and $K = p + m$.

- The parameter of interest is given by the following collection of matrices

$$\beta_k^{(l)} = \begin{pmatrix} \beta_{n_l,1,k}^{(l)} & \cdots & \beta_{n_l,d,k}^{(l)} \\ \vdots & \ddots & \vdots \\ \beta_{n_{l+1},1,k}^{(l)} & \cdots & \beta_{n_{l+1},d,k}^{(l)} \end{pmatrix},$$

where $k = 1, \dots, K$ and $l = 1, \dots, L$.

Consequences & Extensions

Multi-Trigger Systems

- In practice, multiple **competing incidents** can contribute to trigger in the follow-up event.
- To address this we define a **multi-trigger system** where we consider $\mathcal{Y}_1, \dots, \mathcal{Y}_K$ as a sequence of identically distributed random fields with

$$\mathcal{Y}_k = \{Y_{k,\mathbf{x}} : \mathbf{x} \in \mathcal{X}\}, \quad k = 1, \dots, K.$$

- Our framework **extends to the framework of K trigger events** by considering

$$\alpha_I(\mathbf{x}) = \lim_{u \rightarrow y^*} P(I_{u,\mathbf{x}} = 1 \mid Y_{1,\mathbf{x}} > u \vee \dots \vee Y_{K,\mathbf{x}} > u).$$

- This formula simplifies to the original **single-trigger case** by defining $Y_{\mathbf{x}} = \min\{Y_{1,\mathbf{x}}, \dots, Y_{K,\mathbf{x}}\}$, and hence the theory and methods discussed earlier readily apply to this context as well.

Consequences & Extensions

Categorical, Ordinal, and Continuous Follow-up Events

- In real-world applications, the follow-up extreme event may come in different flavors or categories.

Example

For example, $j = 0$ may represent *no tornado*, $j = 1$, *supercell tornado*, and $j = 2$ a *non-supercell tornado*.

- The proposed cascading probability surfaces naturally extend to this context as follows

$$\alpha_I(\mathbf{x})^{(j)} = \lim_{u \rightarrow y^*} P(I_{u,\mathbf{x}} = j \mid Y_{\mathbf{x}} > u),$$

where $j = 0, \dots, J$.

Theoretical Properties

The Kolmogorov Superposition Operator

Highlight

As shown below small perturbations in the inner and outer functions induce only small perturbations in the resulting function.

- Throughout, $\|f\|_\infty \equiv \|f\|_\infty^A := \sup_{x \in A} |f(x)|$, and $C(A)$ and $C_{\text{Lip}}(A)$ denote the spaces of continuous and Lipschitz continuous functions on $A \subseteq \mathbb{R}$.
- Let $I = \{1, \dots, 2d + 1\}$ and $J = \{1, \dots, d\}$.
- Finally, we equip $C([0, 1])^{(2d+1)d} \times C_{\text{Lip}}(\mathbb{R})^{2d+1}$ with the max norm

$$\|\Phi\| = \|(\Phi^{(1)}, \Phi^{(2)})\| = \max(m_1, m_2),$$

where

$$m_1 = \max_{(i,j) \in I \times J} \|\Phi_{ij}^{(1)}\|_\infty, \quad m_2 = \max_{i \in I} \|\Phi_i^{(2)}\|_\infty.$$

Theoretical Properties

Continuity of Kolmogorov Superposition Operator

Theorem

Consider the operator

$$\mathcal{K} : C([0, 1])^{d(2d+1)} \times C_{\text{Lip}}(\mathbb{R})^{2d+1} \rightarrow C([0, 1]^d),$$

defined by $\mathcal{K}(\Phi^{(1)}, \Phi^{(2)})(x_1, \dots, x_d) = \sum_{i=1}^{2d+1} \Phi_i^{(2)}(\sum_{j=1}^d \Phi_{i,j}^{(1)}(x_j))$, with

$$\Phi^{(1)} = (\Phi_{i,j}^{(1)})_{(i,j) \in I \times J} \in C([0, 1])^{d(2d+1)}, \quad \Phi^{(2)} = (\Phi_i^{(2)})_{i \in I} \in C_{\text{Lip}}(\mathbb{R})^{2d+1},$$

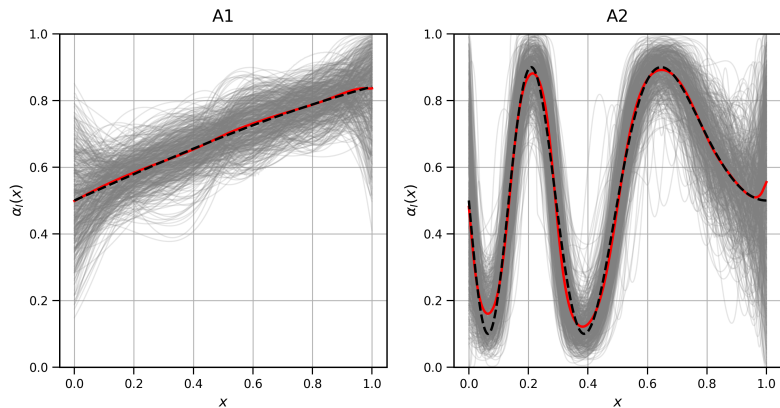
and $I = \{1, \dots, 2d+1\}$ and $J = \{1, \dots, d\}$. Then, $\mathcal{K}$ is a continuous operator.

Monte Carlo Simulation Study

Scenarios A

Scenarios A1 and A2 (n = 5 000)

— MC Mean KAN POC Curve - - - True POC Curve

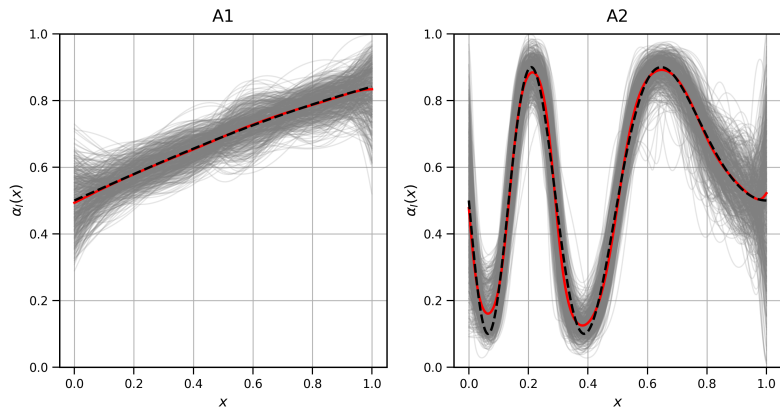


Monte Carlo Simulation Study

Scenarios A

Scenarios A1 and A2 (n = 10 000)

— MC Mean KAN POC Curve - - - True POC Curve

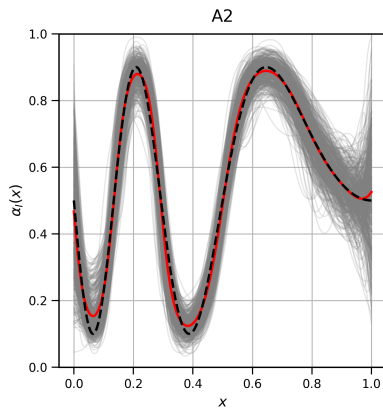
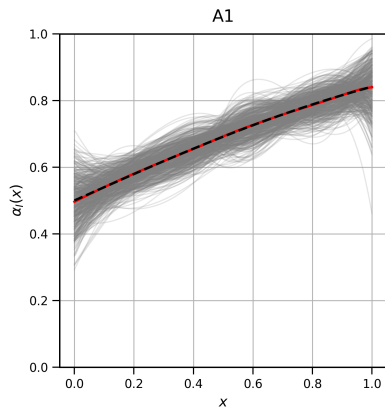


Monte Carlo Simulation Study

Scenarios A

Scenarios A1 and A2 (n = 15 000)

— MC Mean KAN POC Curve - - - True POC Curve



Real Data Illustration

Applied Rationale and Data Description

- Coastal regions are highly vulnerable to [tsunamis](#), with their impacts often compounded by extreme earthquakes that act as primary [triggers](#).
- We now apply the proposed method to quantify the [probability of cascade](#) for tsunami occurrence triggered by extreme earthquakes.
- Our analysis uses data from the [NCEI/WDS Global Significant Earthquake Database](#), provided by the [NOAA National Centers for Environmental Information](#).
- The dataset contains over 5 700 significant earthquakes from 2150 B.C. to present, defined by criteria such as fatalities, damages over \$1 million, Modified Mercalli Intensity (MMI) X or greater, or the earthquake generated a tsunami.
- Each observation includes event date, location, depth, magnitude, MMI, and socio-economic impacts (casualties, injuries, property damage), with references and notes on related events like tsunamis and eruptions.

Real Data Illustration

Implementation

- We threshold the data at their 95% threshold for fitting the POC surface and consider the features,

(latitude, longitude, depth).

- We transform longitude and latitude coordinates to the unit square for modeling purposes.
- Finally, we fit a KAN model with three layers, using a sigmoid activation function at the output.

Real Data Illustration

Visualization

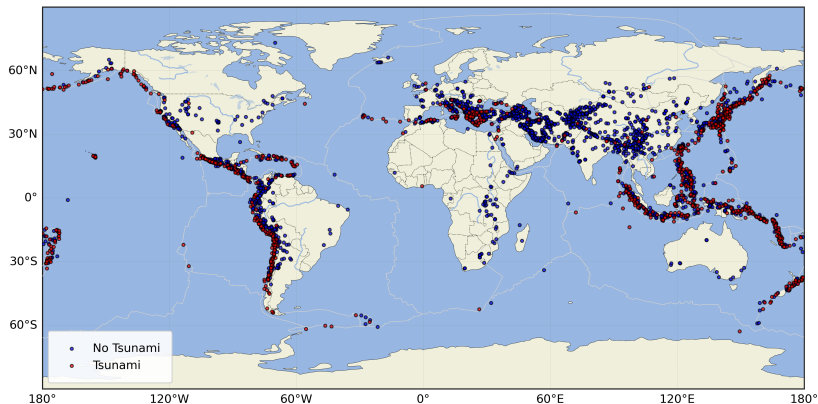
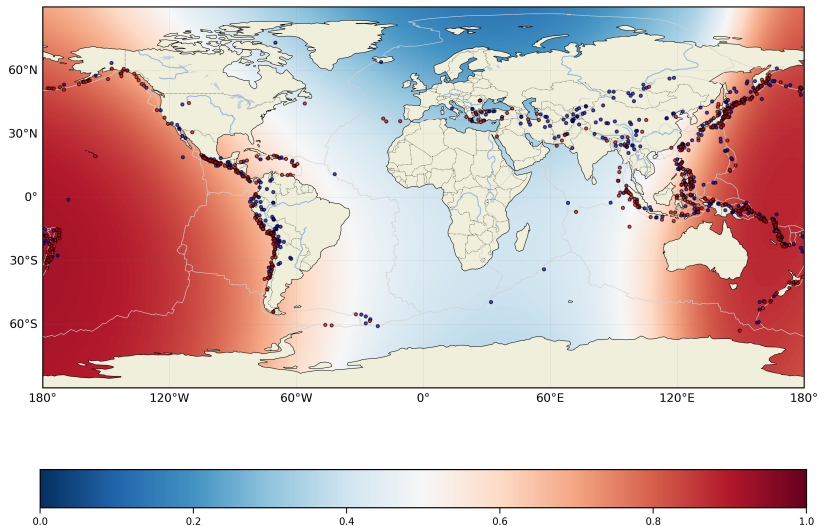


Figure: Point pattern of earthquakes (red) and associated tsunami occurrences (blue).

Real Data Illustration

POC Surface—Depth: Percentile 5



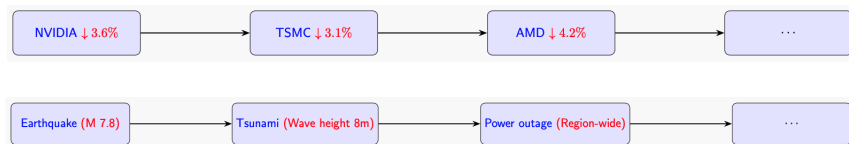
Part II

Generative Transformer-Based Approaches for Cascading Extremes

Rationale

Cascades are Natural Models for Generative Extremes

- Chains of extreme events and cascades of extreme events as proposed in [de Carvalho et al.2026](#)) offer a natural starting point for thinking about generative extremes.



Context

Attention is All you Need

- The starting point for the construction of our first generative AI approach for extreme events is based on notions of [multi-headed attention](#) as well as of [transformers](#), as introduced in [Vaswani et al \(2017\)](#); for an introduction see [Bishop & Bishop \(2023; §12\)](#) as well as Lecture from Day 1.

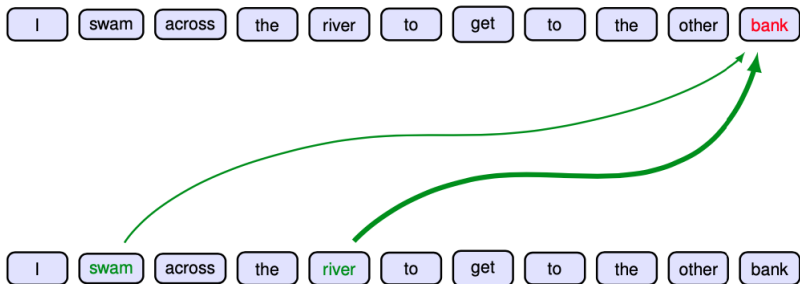


Figure 12.1 Schematic illustration of attention in which the interpretation of the word 'bank' is influenced by the words 'river' and 'swam', with the thickness of each line being indicative of the strength of its influence.

Extremes as Tokens

From Context to Conditional Generation

- Whereas the unit of analysis in a standard **transformer** is a sequence of **tokens**, our EVT-based transformer operates on a **sequence of extreme events**.



- By analogy with language models, where the meaning of a token depends on its context, the interpretation of an extreme event similarly depends on the surrounding events.

Example (Semiconductor losses)

The sequence of extreme stock losses shown in the above figure is more indicative of a tech-specific cascade than of a broader macro-financial shock.

Learning targets: the **most probable continuation** of an extreme-event sequence and the **conditional probabilities** of subsequent extremes.

Most Probable Continuation

Details

- The **most probable continuation** is the mode of the **predictive distribution**

$$\mathbf{x}_{n+1}^* = \arg \max_{\mathbf{x}} p(\mathbf{x} \mid \mathbf{x}_1, \dots, \mathbf{x}_n).$$

This is tantamount to greedy search (Bishop & Bishop, 2023).

- Evidently, this is not the same as the **most probable sequence**, which is given by maximizing the **joint distribution**

$$(\mathbf{x}_1^*, \dots, \mathbf{x}_n^*) = \arg \max_{(\mathbf{x}_1, \dots, \mathbf{x}_n)} p(\mathbf{x}_1, \dots, \mathbf{x}_n).$$

On Extremal Tokens

Two Routes to Generative Cascades

Two research projects we are currently exploring:

- Project 1: “Tokens of marginal exceedances.”

Lee, Richards, Sabanis

Categorical token process; pre-trained transformer and amortized inference.

- Project 2: “Tokens of radial exceedances.”

Ferrer, Vallejos, Ma, Sabanis

Marked Hawkes process and multivariate extreme value theory.

Aldous’s interacting-particle viewpoint motivates a useful contrast: a **conserved token** versus a **state that spreads**.

Bernoulli 19(4), 2013, 1122–1149
DOI: 10.1017/S0005300113000046

Interacting particle systems as stochastic social dynamics

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The style of mathematical models known to probabilists as Interacting Particle Systems and exemplified by the Voter, Exclusion and Contact processes have found use in many scientific disciplines. In many such disciplines the underlying conceptual picture is of a social network, where individuals meet pairwise and update their “state” (opinion, activity etc.) in a way depending on the two pairwise states. This picture motivates a generic generalization to all Finite Markov Decision Exchanges (FMDEs). We briefly describe a few less familiar models (Arranging, Compulsive Gambler, Diffusion, Endonucleic) suggested by the social network picture, as well as a few familiar ones.

Keywords: opinionistic, interacting particle system, Markov chain, social network, voter model

1. Introduction

What is the most broad-ranging currently active field of applied probability? Measuring breadth by the number of different academic disciplines where it appears, it seems hard to beat a field appearing in Physics, Computer Science and Electrical Engineering, Economics and Finance, Psychology and Sociology, Population Genetics and Epidemiology and Ecology. Curiously, the field has no good name, though members of Bernoulli will most likely know it, under the continuing influence of Liggett’s 1985 book [17], as *Interacting Particle Systems*. It turns out that mathematically similar toy models of different real-world entities have been repeatedly re-invented in the different disciplines mentioned above, and literally thousands of papers have been written since 2000 on discipline-dependent variant models invented as more realistic.

Rather than attempting a brief overview of the whole field as it stands today (on which I am certainly not an expert), let me take an appetizer approach, imagining starting with a blank slate. I will not set out the more technical conceptual setting – what sorts of things one may try to model – then a particular technical setup, and then indicate what sort of mathematical results one might seek. Given this motivation, the rest of the article describes some basic models, gives pointers to the existing literature and presents a few open problems suggested by our setting. The article is based on a short course given in summer 2012 at Warwick and Cornell, and more extensive informal material is available at [15]. I must emphasize that the article is intended as an introduction to the topics it mentions, not as an authoritative survey, which would require ten times the length. In particular some of the mathematics outlined here is both new and non-standard, except perhaps some open problems.

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D. Aldous, *Bernoulli* 19(4), 2013.

Finite Markov Information Exchange Processes

Meeting Geometry and Information Updates

- Let $\text{Agents} = \{1, \dots, n\}$ and let (ν_{ij}) be a nonnegative irreducible array indexed by unordered pairs $\{i, j\}$.
- Each pair with $\nu_{ij} > 0$ meets at the times of an independent rate- ν_{ij} Poisson process. This collection is the **meeting process**.
- Let States be a state space and write $X_i(t)$ for the state of agent i at time t .

Definition (FMIE process (Aldous, 2013, §2.2))

States change only at meetings and the update depends only on the two pre-meeting states. For a deterministic rule $F : \text{States}^2 \rightarrow \text{States}$,

$$(s_1, s_2) \mapsto (F(s_1, s_2), F(s_2, s_1)).$$

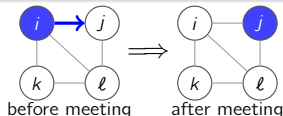
Two ingredients: (ν_{ij}) specifies the geometry; F specifies how information is exchanged.

Two Basic FMIE Processes

The Geometry is Fixed; the Update Rule Changes

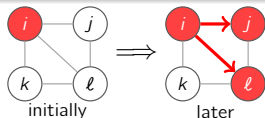
Example (Token process)

There is one token. When the agent i holding it meets agent j , the token passes to j .



Example (Pandemic process)

Initially one agent is infected. Whenever an infected agent meets an uninfected agent, the latter becomes infected.



- **Analogy for extremes:** the Token process motivates one selected label at a time, whereas the Pandemic process motivates branching. The models below are analogues, not the same dynamics.

Project 1: Extremal Vocabulary

Thresholding Multivariate Observations

- Let $\{\mathbf{Y}_t\}_{t=1}^N$ be a d -dimensional process, with $\mathbf{Y}_t = (Y_{t,1}, \dots, Y_{t,d})^\top$.
- Let F_j be the marginal distribution function of component j . At a common high probability level q , set

$$u_j = F_j^{-1}(q), \quad j = 1, \dots, d.$$

- Component j is extreme at time t when $Y_{t,j} > u_j$. Write

$$I_{t,j} = I(Y_{t,j} > u_j), \quad \mathcal{E}_t = \{j : I_{t,j} = 1\}.$$

Definition (Extremal vocabulary)

$\mathcal{V} = \{0, 1, \dots, d\}$, where 0 denotes the **quiet state** and token j identifies component j as the dominant active extreme.

Joint work with J. Lee, J. Richards, and S. Sabanis.

Project 1: Extremal Tokens

Dominant Exceedances on a Common Probability Scale

Definition (Active extremal token)

The multivariate observation $\mathbf{Y}_t$ is mapped to one token $A_t \in \mathcal{V}$:

$$A_t = \begin{cases} 0, & \mathcal{E}_t = \emptyset, \\ \arg \max_{j \in \mathcal{E}_t} F_j(Y_{t,j}), & \mathcal{E}_t \neq \emptyset. \end{cases}$$

- The probability scale makes heterogeneous margins comparable and selects one active component when several exceed simultaneously.
- The token retains the selected component and temporal order, but discards magnitude and non-dominant simultaneous exceedances.

Example (Extremal sentence)

$$\mathcal{S}^{(\psi)} = (A_\tau, \dots, A_{\tau+N_\psi-1}, 0), \quad A_s > 0 \text{ for } \tau \leq s < \tau + N_\psi.$$

The final quiet token 0 delimits the cascade.

Project 2: Setup

Introduction to Geometric Extremes

- Let $\{\mathbf{Y}_t\} = \{(Y_{1,t}, \dots, Y_{d,t})^\top\}$, be a d -dimensional stochastic process with standard Laplace margins.
- Under regularity conditions (Nolde and Wadsworth2022; Wadsworth and Campbell2024; Murphy-Bartrop et al.2025) the joint density satisfies

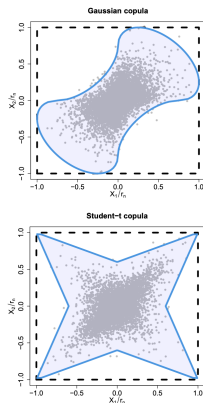
$$-\log f_{\mathbf{Y}_t}(u\mathbf{x}) \sim u g_t(\mathbf{x}), \quad \text{as } u \rightarrow \infty,$$

where the **gauge function** $g_t : \mathbb{R}^d \rightarrow \mathbb{R}_+$ is continuous and homogeneous, i.e., for any $c > 0$

$$g_t(c\mathbf{w}) = c g_t(\mathbf{w}).$$

- The sublevel set

$$G_t := \{\mathbf{x} \in \mathbb{R}^d : g_t(\mathbf{x}) \leq 1\}$$



Setup

Magnitude and Direction of Extremes

- Write $\mathbf{X}_t = R_t \mathbf{W}_t$ with

$$R_t = \|\mathbf{X}_t\|, \quad \mathbf{W}_t = \mathbf{X}_t / R_t \in \mathbb{S}^{d-1}.$$

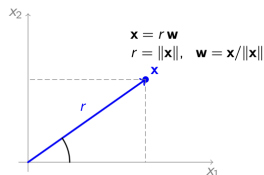
- On $\{R_t > u_t(\mathbf{w})\}$, the density of $\mathbf{X}_t$ factorizes

$$f_t(\mathbf{x}) = f_{R_t|\mathbf{W}_t}(r \mid \mathbf{w}) h_t(\mathbf{w}), \quad \mathbf{x} = r\mathbf{w}.$$

- Here, h_t is the angular surface on $\mathbb{S}^{d-1}$ (Castro et al.2018) and $f_{R_t|\mathbf{W}_t}$ is a truncated gamma density (Murphy-Barltrop et al.2025):

$$f_{R_t|\mathbf{W}_t}(r \mid \mathbf{w}) \propto \{g_t(\mathbf{w})\}^d r^{d-1} \exp\{-rg_t(\mathbf{w})\},$$

for $r > u_t(w)$.



Time Between Extremes

- Time between exceedances is modelled using a (marked) Hawkes process.
- Each exceedance temporarily raises the intensity (self-exciting).
- The marks take values in $\mathbb{S}^{d-1} \times \mathbb{R}_+$.
- The exceedance times $T_1 < T_2 < \dots$ and marks $(\mathbf{W}_i, R_i)$ satisfying $R_i > u_i$ form a marked point process

$$N = \sum_{i \geq 1} \delta_{(T_i, \mathbf{W}_i, R_i)},$$

with intensity

$$\lambda_t = \mu_t + \sum_{T_j < t} \psi_j(t).$$

- Here $\mu_t > 0$ is the baseline intensity and $\psi_j(t) \geq 0$ is the excitation kernel associated with event j , with $\psi_j(t) = 0$ for $t \leq T_j$ and $\int_{T_j}^{\infty} \psi_j(t) dt < \infty$.

Cascading Extremes

- For each event i , the **parent variable** $P_i \in \{0, 1, \dots, i-1\}$ has distribution

$$P(P_i = 0) = \frac{\mu_{T_i}}{\lambda_{T_i}}, \quad P(P_i = j) = \frac{\psi_j(T_i)}{\lambda_{T_i}}, \quad 1 \leq j < i.$$

- An event with $P_i = 0$ is an **immigrant**; an event with $P_i = j \geq 1$ is an **offspring** of event j .

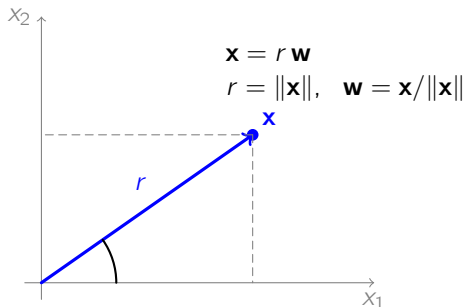
Definition (Cascade)

A cascade $\mathcal{C}$ consists of an immigrant i_0 and all its descendants under the parent relation.

- Since $P_i < i$, each cascade is a rooted tree.
- The offspring set of j is

$$\nabla_j(\mathcal{C}) = \{i \in \mathcal{C} : P_i = j\}.$$

Summary of Model



- Angular component: Mixture of von Mises.
- Radial component: Truncated gamma ([Wadsworth–Campbell](#)).
- Time between exceedances: Hawkes process.

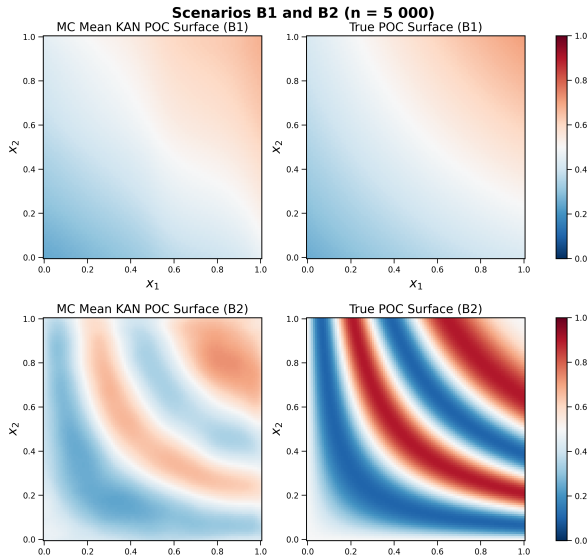
Closing Remarks

Summary

- This talk presented a novel statistical framework for [cascading extreme events](#)—such as earthquakes triggering tsunamis or heatwaves sparking wildfires, with further downstream losses.
- The proposed approach aims to offer a novel outlook into [triggering extremal events](#) and their [domino effects](#).
- [KANE](#), a neural model based on [Kolmogorov's superposition theorem](#), was developed to learn about the proposed [POC surface](#).
- In addition, I offered some remarks on how cascades offer a natural starting point for thinking about [generative extremes](#).
- Our EVT-based transformer represents the multivariate extremal process as a [sequence of categorical tokens](#).

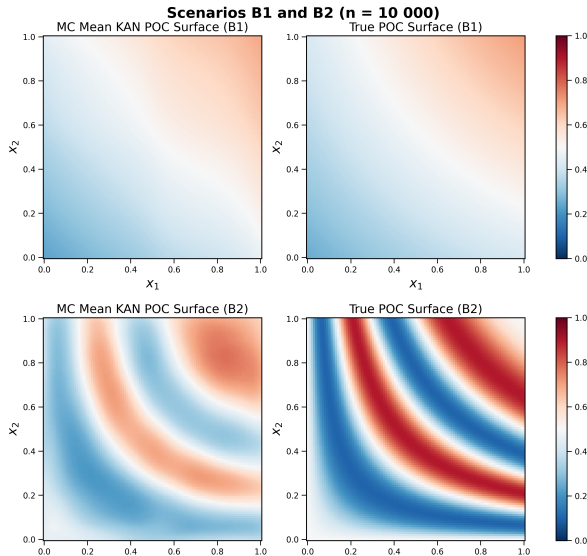
Closing Remarks

Scenarios B



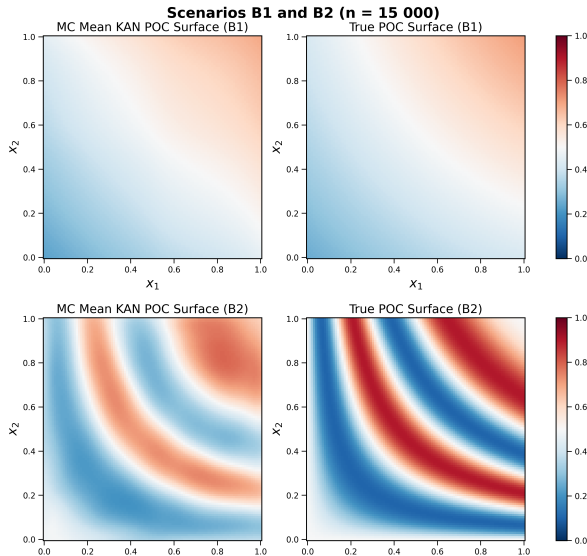
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Scenarios B



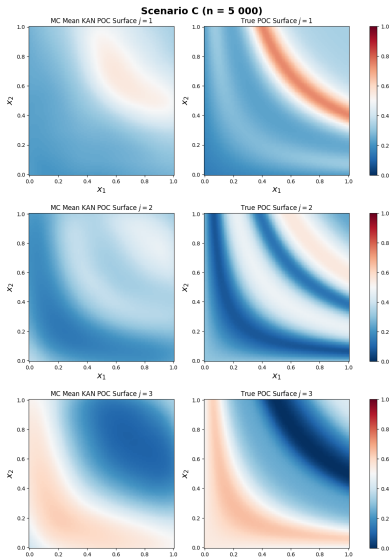
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Scenarios B



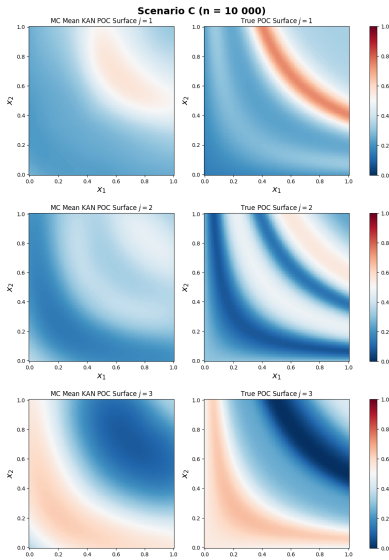
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Scenarios C



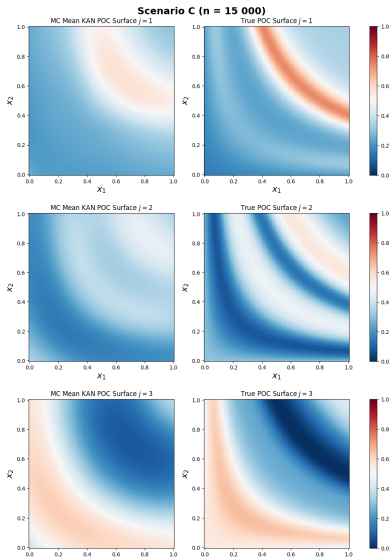
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Scenarios C



Closing Remarks

Scenarios C



On the Continuity of POC

Theorem

Let $\{(l_{\mathbf{x},u}, Y_{\mathbf{x}}) : \mathbf{x} \in [0, 1]^d, u > 0\}$. Assume:

- (a) The mapping $\mathbf{x} \mapsto P(l_{\mathbf{x},u} = 1 \mid Y_{\mathbf{x}} > u)$ is continuous on $[0, 1]^d$, for all $u \in \mathbb{R}$;
- (b) The limit $\lim_{u \rightarrow y^*} P(l_{\mathbf{x},u} = 1 \mid Y_{\mathbf{x}} > u)$ exists and convergence is uniform in $\mathbf{x}$ over $[0, 1]^d$.

Then, $\mathbf{x} \mapsto \alpha(\mathbf{x}) \equiv \lim_{u \rightarrow y^*} P(l_{\mathbf{x},u} = 1 \mid Y_{\mathbf{x}} > u)$ is continuous on $[0, 1]^d$.

Universal Approximation Theorem

Theorem (Universal Approximation Theorem)

Suppose f is a continuous function on a compact space $\mathcal{X} \subset \mathbb{R}^d$ and σ is not a polynomial. Then, for any $\varepsilon > 0$, there exists a **one-hidden layer** neural network f_{θ} such that

$$\sup_{\mathbf{x} \in \mathcal{X}} |f(\mathbf{x}) - f_{\theta}(\mathbf{x})| < \varepsilon.$$

Notation: Here $f_{\theta} : \mathbb{R}^d \rightarrow \mathbb{R}$ is a one-hidden layer feedforward neural network composed of K neurons

$$f_{\theta}(\mathbf{x}) = b^{(2)} + \sum_{k=1}^K w_k^{(2)} \sigma(\langle \mathbf{w}_k^{(1)}, \mathbf{x} \rangle + b_k^{(1)}),$$

where

$$\theta := \{b^{(2)}\} \cup \{\mathbf{w}_k^{(1)}, w_k^{(2)}, b_k^{(1)}\}_{k=1}^K \in \Theta := \mathbb{R} \times (\mathbb{R}^d \times \mathbb{R} \times \mathbb{R})^K,$$

and where $\langle \cdot, \cdot \rangle$ is a scalar product on $\mathbb{R}^d$, and $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is a non-linear activation function.

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